

Erdős Problem #1002

EP-1002	L1 - Tractable	Ouvert	Combinatorics
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Énoncé original (anglais, reproduction mot pour mot)

For any $0 < \alpha < 1$, let $f(\alpha, n) = \frac{1}{\log n} \sum_{1 \leq k \leq n} \left(\frac{1}{2^k} - \alpha^k \right)$.

Does $f(\alpha, n)$ have an asymptotic distribution function?

In other words, is there a non-decreasing function g such that $g(-\infty) = 0$, $g(\infty) = 1$,

and $\lim_{n \rightarrow \infty} \left| \left\{ \alpha \in (0, 1) : f(\alpha, n) \leq c \right\} \right| = g(c)$?

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